

A Calibration Method of Portable Coordinate Measuring Arms by using Artifacts

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Received: 26 March 2018 / Accepted: 08 November 2018

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Abstract: Portable coordinate measuring arm (PCMA) calibration is an efficient way to enhance PCMA accuracy. Optimal test configurations during PCMA calibration are helpful to reduce measuring points and reduce consumption time for calibration. In this paper, a kinematic calibration approach has been proposed for 5DOF PCMA, in which ASME B89.4.22-2004 standard and a self-made ball bar for volumetric calibration test are employed. First, a kinematic error model of PCMA is established, and the error parameters are identified. Second, data processing is carried out based on data obtained from tests including single-point precision calibration test using chamfered hole seat, effective diameter calibration test using the reference spheres, volumetric and distance displacement calibration test using ball bar, and machine coordinate origin determination test using the base plane and base cylinder. Third, the objective function for the identification of kinematic error parameters is established, and it is solved by using Levenberg–Marquardt algorithm. Finally, the results of calibration test are compared with the ANSI/ASME B89.4-2004 performance test. The experimental results show that the single-point repeatability of the machine is reduced 1.1 times and the distance accuracy is enhanced 1.17 times by data processing.

Keywords: AACMM; Kinematic error model; Repeatability; Measurement accuracy; Calibration

1. Introduction

An articulated arm coordinate measuring machine (AACMM) is the new type of mobile measuring equipment. Recently, many researchers try to improve the performance of AACMM, and some progress has been made. The structure parameters of AACMM should be reasonably settled during the design considering the factors such as the circuit installation, the size of rotary encoder, and the measuring probe style. To enhance the accuracy of AACMM, kinematic error calibration is necessary. Kinematic calibration is the process of identifying kinematic parameters of the machine such as angle offsets and link lengths [1]. A kinematic calibration procedure normally

consists of five steps: determination of the kinematic model, data acquisition, parameter identification, validation, and finally, implementation of correction models [2]. This approach focuses on the structure parameter identification based on the error model [3–14].

In general, kinematic calibration is performed using two nonlinear least squares optimization methods, one of which is the system theory method [15] and the other is a geometrically significant recognition method, in which several measurement and calculation steps are employed [16, 17]. The advantage of the model-based method is that the measuring space can be accurately calibrated and all position and orientation errors within the calibrated range can be compensated from all joint angles [18].

Calibration of AACMMs is often used by users to verify coordinate measuring machines (CMM) error [19]. Performance tests of AACMM are carried frequently out by

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using various artifacts such as ball bar (a carbon fiber profile and 15 ceramic spheres of 22 mm in diameter at 1.5 m long), ball plate with size 400×400 mm, virtual spheres gauge, virtual spheres plate with 16 groups of four conic holes fixed at a $300 \text{ mm} \times 300 \text{ mm} \times 20 \text{ mm}$ aluminum plate, hole plate consisting of 25 holes in a 5×5 square pattern with a nominal pitch of 20 mm, and a hybrid gauge having a virtual ball bar with four holes at ends and a ball bar [20–26].

Especially, the ASME B89.4.22-2004 standard pointed out methods for performance test of AACMMs called as effective diameter, single-point articulation and volumetric performance test [27].

This work introduces methods to test the performance of self-designed and produced PCMA. (This device is used professionally to inspect automobile.) For the calibration, ASME B89.4.22-2004 standard is referenced and a self-made 1.4-m-long ball bar having 8 spheres is used for the volumetric performance test. A new method of data processing for the exclusion of rough errors is presented, and the machine coordinate origin determination is carried using the base plane and base cylinder. An objective function for the identification of kinematic error parameters is derived and solved by using Levenberg–Marquardt algorithm (LMA). An experiment for the verification is conducted, and the accuracy of the presented calibration method is validated.

2. Kinematic Model of the 5-DOF PCMA

The manufactured structure of the 5-DOF PCMA is shown in Fig. 1. The machine has a five-axis revolute joint. Figure 2 shows the reference frames of the 5 links and end effector, and some of DH parameters. The reference



Fig. 1 Photograph of the 5DOF PCMA

coordinate system is established by employing the modified DH (MDH) model with y-axis twist.

The structure of the PCMA consists of the joints whose rotation axes either parallel or vertical. More precisely, the machine has 2 pairs of parallel consecutive joints ($z_1 \parallel z_2$, $z_2 \parallel z_3$) and has 2 pairs of consecutive vertical joints ($s_i, \bar{s}$). As pointed out by [28, 29], when two consecutive joints are parallel or nearly parallel, a small variation in link parameters could be cause of enormous change in the Denavit–Hartenberg (DH) parameters. Thus, we employ the modified DH (MDH) model with y-axis twist [30].

The MDH matrix, f_{point} , which performs the transformation from the s_i th frame to the f_{sphere} th frame, is given as follows:

$$T_i^{i-1} = \text{Rot}(z, \theta_i) \text{Trans}(z, d_i) \text{Rot}(x, \alpha_i) \text{Trans}(x, a_i) \text{Rot}(y, \beta_i)$$

$$= \begin{pmatrix} c\theta_i c\beta_i - s\alpha_i s\theta_i s\beta_i & c\alpha_i s\theta_i & c\theta_i s\beta_i + s\alpha_i s\theta_i c\beta_i & a_i c\theta_i \\ s\theta_i c\beta_i + s\alpha_i c\theta_i s\beta_i & c\alpha_i c\theta_i & s\theta_i s\beta_i - s\alpha_i c\theta_i c\beta_i & a_i s\theta_i \\ -c\alpha_i s\beta_i & s\alpha_i & c\alpha_i c\beta_i & d_i \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad (2.1)$$

where the parameters of link D_{max} include joint rotation angle f_{length} , link offset $L_i - L_{\text{cal}}$, link twist angle $z = 0$, link length $f_{z=0}$, and link twist angle about the z_i axis $x^2 + y^2 = R^2$. Apparently, MDH matrix has one more parameter than the DH matrix. But link offset f_{cylinder} or link twist angle about the $\text{cyl} \delta_i$ axis β_i is exactly zero which depends on whether two consecutive joints are parallel or not. $d_i = 0$ in the case of consecutive joint axis Z_{i-1} and Z_i are (nearly) parallel, and $\beta_i = 0$ in otherwise. Thus, each joint actually has 4 MDH parameters (in the case of parallel joint, $\theta_i, \alpha_i, a_i, \beta_i$; and otherwise $\theta_i, d_i, \alpha_i, a_i$). The nominal MDH parameters of the PCMA are listed in Table 1. In the table, notation “-” means that there is not the parameter.

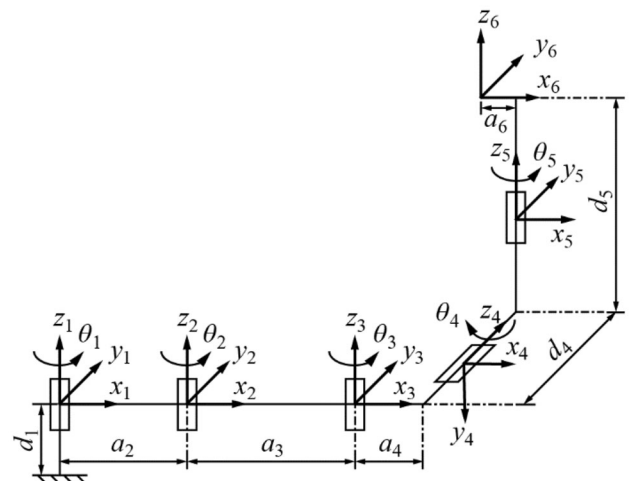


Fig. 2 Reference frames

Given the configuration vector $\theta_i = (\theta_1, \theta_2, \theta_3, \theta_4, \theta_5)^T$ of the joint variables, the probe's pose is represented as follows:

$$A = T_1^0 T_2^1 T_3^2 T_4^3 T_5^4 \quad (2.2)$$

Finally, the coordinates of the probe center with respect to the base frame $p = [x, y, z, 1]^T$ are represented as follows:

$$p = \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = A \cdot \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \quad (2.3)$$

The kinematic model of PCMA has 21 parameters. Some of them, however, are not identifiable, and they could be determined by using a technique from [31]. Among the 21 parameters of the PCMA model that are listed in Table 1, θ_1 and d_1 are not identifiable, α_5 and a_4 are the coupling parameters of α_4 and a_5 , respectively.

The coupling parameters cannot be determined together. Finally, we have a total of 17 error parameters of the PCMA to identify in our calibration process (see Table 2). In Table 2, notation “ $\times$ ” indicates the parameters that are not identifiable.

The kinematic model of the PCMA that includes the error parameters can be represented as follows:

$$p = \text{Trans}(\theta_i; \Delta) \quad (2.4)$$

where *Trans* stands for the transformation relation in Eq. (2.3), $\theta_i = (\theta_1, \theta_2, \theta_3, \theta_4, \theta_5)^T$ is a vector of joint configuration, and $\Delta = (\delta\theta_2, \delta\theta_3, \delta\theta_4, \delta\theta_5, \delta d_4, \delta d_5, \delta\alpha_1, \delta\alpha_2, \delta\alpha_3, \delta\alpha_4, \delta a_1, \delta a_2, \delta a_3, \delta a_5, \delta\beta_1, \delta\beta_2, \delta\beta_3)^T$ is the error parameter vector that should be identified. Here, a

Table 1 Nominal MDH parameters (notation “—” means there is not the parameter)

No.	θ_i (deg)	d_i (mm)	α_i (deg)	a_i (mm)	β_i (deg)	Range of θ_i (deg)
1	θ_1	156.5	0	0	0	[− 175, 175]
2	θ_2	—	0	420	0	[− 160, 160]
3	θ_3	—	0	450	0	[− 130, 130]
4	θ_4	− 65.5	− 90	85	—	[− 45, 135]
5	θ_5	682	90	− 118.4	—	[0, 360]

Table 2 The MDH parameters, including error parameters

No.	θ_i (deg)	d_i (mm)	α_i (deg)	a_i (mm)	β_i (deg)
1	$\theta_1(\times)$	156.5 ($\times$)	$\delta\alpha_1$	δa_1	$\delta\beta_1$
2	$\theta_2 + \delta\theta_2$	—	$\delta\alpha_2$	$420 + \delta a_2$	$\delta\beta_2$
3	$\theta_3 + \delta\theta_3$	—	$\delta\alpha_3$	$450 + \delta a_3$	$\delta\beta_3$
4	$\theta_4 + \delta\theta_4$	− 65.5 + δd_4	− 90 + $\delta\alpha_4$	85 ($\times$)	—
5	$\theta_5 + \delta\theta_5$	682 + δd_5	90 ($\times$)	− 118.4 + δa_5	—

kinematic error parameter $\delta\theta_i$ is the difference between the marked zero position of each encoder and the initial position defined in the kinematic error model. Thus, the joint variable θ_i of the error model is related to each encoder angle reading from the encoder through Eq. (2.5).

$$\theta_i = \theta_i^{\text{Enc}} - \theta_i^0 \quad (2.5)$$

where θ_i^{Enc} : the angle reading from the encoder; θ_i^0 : the marked zero position of encoder.

3. Data Capture, Analysis and Process

The data capture setup consists of acquiring the nominal coordinates of points in the working volume of PCMA by using various geometric artifacts. In this step, there are several issues that must be noted. First of all, the positions of the “captured” points should cover the whole working volume of PCMA as much as possible. There are some unmodeled sources of error such as the elasticity of links and the backlash of joints that will affect the parameter identification. As a result, the positioning accuracy far from “captured” working volume will be worse than that of in the “captured” working volume. Secondly, the rotation angles of joints during data capture setup should cover the whole rotation range of joints as much as possible. Finally, the “captured” points should guarantee that the geometric parameters of artifacts could be identified from that point.

In the present work, we use various geometric artifacts: the chamfered hole seat for single-point precision calibration, the precision reference sphere for the effective diameter performance test and ball bars for the linear displacement accuracy and the volumetric performance test. A hard probe is used for all the calibrations.

The error due to the kinematic parameter error is very large in measuring machines that have not yet been calibrated. If the standard operation is not undertaken during calibration, it will not be able to get a more accurate solution due to incoherent numerical data in the process of obtaining optimized solution. For the further improvement in the measurement accuracy of the measuring machine, it is necessary to carry out a recalibration, which requires a lot of labor and time. This problem could be eliminated by reducing rough errors through data processing during the calibration.

3.1. Single-Point Precision Calibration Test Using the Chamfered Hole Seat

3.1.1. Setup and Measurement Procedures

The single-point precision calibration test is intended to assess the repeatability and reproducibility of the PCMA.

The chamfered hole seat consists of 121 holes in a 11×11 square pattern, with a nominal pitch of 25 ± 0.01 mm. The nominal diameter of the holes is 6 mm. The chamfered hole seat is used for this test (see Fig. 3). This is rigidly mounted at 9 different locations in the working volume. During the test, the probe of PCMA is placed in the chamfered hole seat and the PCMA arm is articulated to 12 different orientations, maximizing the possible configurations of the PCMA joints (Fig. 4).

The first location shall be within length of the first arm, centered at the first rotational axis of the PCMA. The second location shall be within a zone defined by the length of the first arm and second arm. The third location shall be within a zone defined by the length of the second arm and the third arm.

3.1.2. Data Analysis and Process

The statistics used to represent the single-point precision calibration are the maximum deviation of the points from a mean value and twice the standard deviation of the point location. Using Eq. (3.1), the three-dimensional deviations (δ_j) are calculated from the average coordinates to each of the corresponding twelve points obtained at the one measurement location [27]:

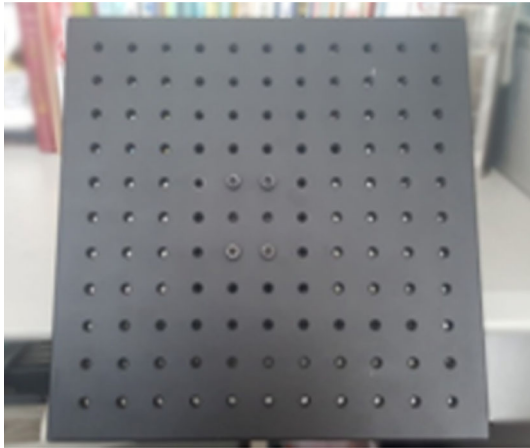


Fig. 3 Chamfered hole seat for single-point precision calibration

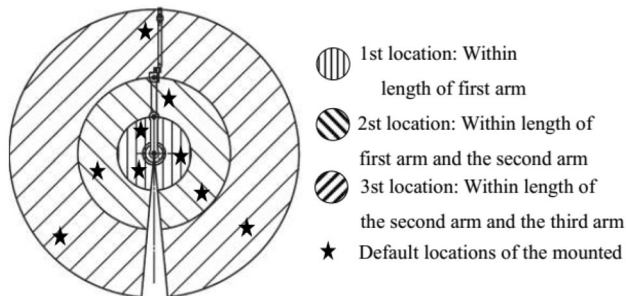


Fig. 4 Nine default locations of the mounted seat during single-point precision calibration

$$\delta_j = \sqrt{(x_j - \bar{x})^2 + (y_j - \bar{y})^2 + (z_j - \bar{z})^2} \quad (3.1)$$

where x_j, y_j, z_j —measured coordinate, $\bar{x}, \bar{y}, \bar{z}$ —average coordinate.

From these values, twice the standard deviation $2s_{pt}$ value of the any one sample is as follows:

$$2s_{pt} = 2\sqrt{\frac{\sum \delta_j^2}{(n-1)}} \quad (3.2)$$

where n = number of measurements (in this case, $n = 12$).

The three-dimensional deviations (δ_{ij}) are calculated corresponding to the whole measurement locations:

$$\text{point } \delta_{ij} = \sqrt{(x_{ij} - \bar{x}_i)^2 + (y_{ij} - \bar{y}_i)^2 + (z_{ij} - \bar{z}_i)^2} \quad (3.3)$$

x_{ij}, y_{ij}, z_{ij} —measured coordinate at the i th measurement location, $\bar{x}_i, \bar{y}_i, \bar{z}_i$ —average coordinate at the i th measurement location.

Therefore, twice the whole standard deviation $2S_{point}$ value is as follows:

$$2S_{point} = 2\frac{s_{pt}}{\sqrt{m}} \quad (3.4)$$

where m = number of measurement locations (in this case, $m = 9$).

If $\text{point } \delta_{ij} > 2S_{point}$, these point data will be removed. These values can be regarded as an error due to a mistake during the measurement for calibration (operator's mistake, for example, hand variation when pressing the measurement button; when not measured at the exact hole location; when the joint arm is not measured in the most stable state). Only when removing these data, we can obtain the better calibration results.

In this work, the calculated results are as follow:

$$\begin{aligned} \text{point } \delta_{ij \max} &= 11.9970 \text{ (mm)} \\ 2S_{point} &= 10.3574 \text{ (mm)} \end{aligned}$$

Find and remove values greater than $2S_{point}$ according to the method described above. The reason measured 12 points at each location is to compensate for these removed values. The modified result is as follows:

$$\begin{aligned} \delta_{ij \max}^{\text{point}} &= 10.2735 \text{ (mm)} \\ 2S_{point} &= 10.3030 \text{ (mm)} \end{aligned}$$

3.2. Effective Diameter Calibration Test using the Reference Sphere

3.2.1. Setup and Measurement Procedures

Effective diameter calibration test is induced to assess the accuracy of the PCMA. The test involves measurements of a reference sphere with nominal diameters ranging between

10 mm and 50 mm [27] and form deviations of $< 1 \mu\text{m}$. The sphere is rigidly mounted at 12 different locations in the working volume (Fig. 4). The sphere diameter is measured using ten points per sphere location and compared with the nominal diameter. During this test, the articulation of the PCMA should be minimized for each measurement.

3.2.2. Data Analysis and Process

The following equation will be used to calculate the radius of a sphere at the one measurement location.

$$r_i = \sqrt{(x_k - \bar{x})^2 + (y_k - \bar{y})^2 + (z_k - \bar{z})^2} \quad (3.5)$$

where r_i —calculated radius, x_k, y_k, z_k —measured coordinate, $\bar{x}, \bar{y}, \bar{z}$ —average coordinate.

The ten points are used to determine a sphere using PCMA software and the diameter calculated. The diameter deviation can be calculated from the difference between the measured and the calibrated diameters of the sphere. The effective diameter calibration test value is reported as the maximum deviation regardless of sign.

$$\text{dia } \delta_i = D_i - D_{\text{cal}} = [2(r_k - r_{\text{probe}})] - D_{\text{cal}} \quad (3.6)$$

where $\text{dia } \delta_i$ —diameter deviation (mm), D_i —measured diameter of sphere (mm), D_{cal} —calibrated diameter of sphere (mm), r_{probe} —radius of the probe (in this case, $r_{\text{probe}} = 4 \text{ mm}$). The results may reveal a number of problems.

3.2.2.1. Selection of the Reasonable Diameter of the Calibrated Sphere

$$D_{\text{cal}} > 2 \cdot \text{point } \delta_{ij \max} \quad (3.7)$$

where $\delta_{ij \max}$ —the maximum deviation of the points during the single-point precision calibration test (mm).

If $D_{\text{cal}} < 2 \cdot \text{point } \delta_{ij \max}$, a reasonable solution calculating the diameter of sphere using the least squares method cannot be obtained. The larger the reference sphere diameter, the smaller the diameter deviation becomes. However, the reference diameter of the sphere cannot be set too large. Also, if the reference diameter of the sphere is too large, the effect is not obvious because the deviation amount is too small. In this work, the effective sphere diameter shall be determined as $D_{\text{cal}} = 50 \text{ mm}$. The points of the diameter deviations shall be removed regardless of sign greater than $\text{point } \delta_{ij \max}$ according to the method described above. Therefore, we would like to recommend first performing the single-point precision calibration and then performing the effective diameter calibration.

3.3. Volumetric Calibration Test Using Ball Bars

3.3.1. Setup and Measurement Procedure

Volumetric calibration test is induced to assess the accuracy of PCMA. In this test, we use a ball bar with 8 spheres that are equally spaced at intended 200-mm interval (see Fig. 5). The spheres have nominal diameter of 30 mm and form deviations of $< 1 \mu\text{m}$. The diameters and the center-to-center distances of the spheres are pre-calibrated by using CMM (Global Advantage 09.20.08 of Hexagon AB with repeatability of $1 \mu\text{m}$ and length accuracy of $2.6 + L/300 \mu\text{m}$).

In the test, we only consider the distances between first and eighth reference spheres and third and sixth spheres. (That is, the nominal lengths of the ball bars are 1400 mm and 600 mm.) Volumetric calibration test at many more locations and orientations would be too time-consuming. Therefore, linear displacement accuracy is combined with the volumetric calibration test. Ball bars are measured in 20



Fig. 5 Ball bar with 8 spheres that are equally spaced at intended 200-mm intervals

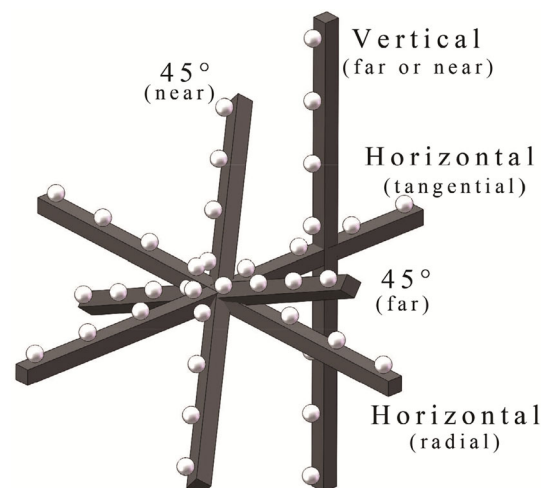


Fig. 6 Ball bar locations for each quadrant

locations: 4 vertical, 8 horizontal, and 8 inclined in 45 deg (Fig. 6).

3.3.2. Data Analysis and Process

For each length measurement, we record seven measurement points per each sphere. The points should be well dispersed on the surface of the sphere [27]. The sphere centers are then used to calculate center-to-center length value L_i for the ball bar position. The length deviations δ_i are calculated from the pre-calibrated ball bar value L_{cal} .

$$\bar{\delta}_i = L_i - L_{cal} \quad (3.8)$$

The results of this test will be described as the maximum deviation, the range of the deviations, and two times the root-mean-square of the deviations. Two times the root-mean-square deviation is calculated from

$$2RMS = 2\sqrt{\frac{\sum \bar{\delta}_i^2}{n}} \quad (3.9)$$

where $n = 20$ in this case.

During test, there appears to be several outlying points that do not conform to the general trend. If the difference between the two length measurements is not within twice the single-point precision calibration, these points will be found and removed as well as the effective diameter calibration.

3.4. Origin Coordinate Calibration test Using Base Plane $z = 0$ and Base Cylinder $x^2 + y^2 = R^2$

3.4.1. Setup and Measurement Process

In order to accurately determine the origin of the base reference frame, it is necessary to calibrate the machine by using the base plane of the machine, $z = 0$, and the cylinder, $x^2 + y^2 = R^2$ (see Fig. 7). If these tests are not con-

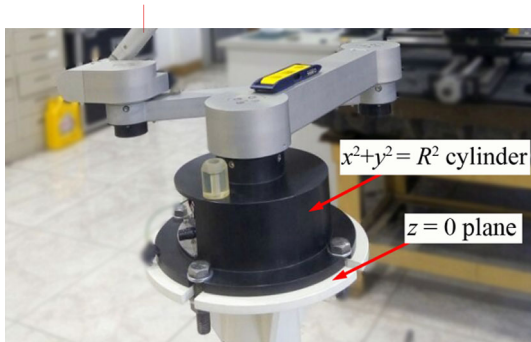


Fig. 7 Base plane $z = 0$ and base cylinder $x^2 + y^2 = R^2$

sidered, then the parameters δa_1 and $\delta \beta_1$ cannot be identified.

During the $z = 0$ plane measuring test, the probe is placed on the base plane and twenty points are captured around the machine on the base plane. During the $x^2 + y^2 = R^2$ cylinder measuring test, the probe is placed on the cylinder surface and twenty points are captured around the machine on the cylinder surface.

Twenty points were measured for the base plane $z = 0$. In order to determine the axis of the z axis, a total of 20 points were obtained by measuring four sections of the cylinder with five points per one section.

3.4.2. Data Analysis and Process

Base plane $z = 0$

The reference plane is obtained by using the least squares method with 20 points of $z = 0$ plane. The distance from the measured points to the determined reference plane is reported as the flatness.

$$d_i = \frac{|Ax_i + By_i + Cz_i + D|}{|A^2 + B^2 + C^2|} \quad (3.10)$$

where $n = (A, B, C)$ —the normal vector of the reference plane (it is obtained using the least squares method), D —the distance required to translate the plane to the origin of the coordinate. (In this case, $D = r_{probe}$.) If $d_i > \text{point} \delta_{ij \max}$, these points will be removed.

The base plane calibration value is reported as the maximum deviation of the flatness.

Base cylinder $x^2 + y^2 = R^2$

The aim of measuring the reference cylinder is to accurately determine the reference axis of the base frame. The twenty points are used to determine a reference axis of the reference cylinder using the least squares method. The distance from the measured points to the determined reference cylinder is reported as the straightness.

$$\text{cyl} \delta_i = R_i - (R_{cal} + r_{probe}) \quad (3.11)$$

where R_i —measured radius $R_i = \sqrt{x_i^2 + y_i^2}$ (mm), R_{cal} —calibrated radius (in this case $R_{cal} = 100$ mm).

If $\text{cyl} \delta_i < \text{point} \delta_{ij \max}$, these points will be found and removed. The base cylinder calibration value is reported as the maximum deviation of straightness.

3.5. Calibration Data and Validation Data

The measurement data are classified into two groups, named *calibration data* and *validation data*. The calibration data are used for identifying the PCMA parameters, and validation data are used for validation of the identified

Table 3 The number of positions for each artifact, and the number of captured points per each position

Artifact	Calibration data		Validation data	
	Locations	Points per location	Locations	Points per location
Chamfered hole	9	12	3	12
Sphere	9	10	3	10
Ball bar	20	7	20	7
$z = 0$ plane	1	20	1	10
Cylinder	1	20	1	10

parameters. The number of locations for each artifact and the number of captured points per each location are listed in Table 3.

4. Calibration Experiment of the PCMA

The kinematic calibration of the PCMA is essentially the procedure to identify the error parameters in Eq. (2.4), and this is achieved by minimization of the objective function, which is expressed in terms of the errors between the nominal parameters and measured one of the geometric artifacts. The minimization problem then could be solved by optimization algorithms such as the Levenberg–Marquardt algorithm (LMA) [32].

4.1. Definition of the Objective Function

4.1.1. Single-Point Precision Calibration Test

The error function for single-point precision calibration test is defined as:

$$f_{\text{point}}(\Delta) = \sum_{i=1}^9 \frac{1}{12} \cdot \text{point} \delta_{ij}^2 = \sum_{i=1}^9 \left[\frac{1}{12} \cdot \sum_{j=1}^{12} [(x_{ij} - \bar{x}_i)^2 + (y_{ij} - \bar{y}_i)^2 + (z_{ij} - \bar{z}_i)^2] \right] \quad (4.1)$$

4.1.2. Effective Diameter Calibration Test

The position and radius of the sphere are determined by using the 12 captured points. Then, the error function for effective diameter calibration test is defined as:

$$f_{\text{sphere}}(\Delta) = \sum_{i=1}^{12} \text{dia} \delta_i^2 \quad (4.2)$$

4.1.3. Volumetric Calibration Test

The error function for volumetric calibration test is then defined as:

$$f_{\text{length}}(\Delta) = \sum_{i=1}^{20} \text{bar} \delta_i^2 \quad (4.3)$$

4.1.4. Base plane $z = 0$ and Base cylinder $x^2 + y^2 = R^2$ calibration test

For the 20 captured points on the $z = 0$ plane, (x_i, y_i, z_i) , $i = 1, 2, \dots, 20$, the error function is defined as:

$$f_{z=0} = \sum_{i=1}^{20} (z_i - r_{\text{probe}})^2 \quad (4.4)$$

where $r_{\text{probe}} (= 4 \text{ mm})$ is the radius of the probe of PCMA.

For the 20 points captured on the surface of $x^2 + y^2 = R^2$ cylinder, (x_i, y_i, z_i) , $i = 1, 2, \dots, 20$, the error function is defined as:

$$f_{\text{cylinder}} = \sum_{i=1}^{20} \text{cyl} \delta_i^2 \quad (4.5)$$

4.1.5. Definition of the Objective Function for Calibration

The objective function should reflect all the error functions of each calibration test. Therefore, the objective function should have the following type:

$$f_{\text{total}} = f_{\text{point}} + f_{\text{sphere}} + f_{\text{length}} + f_{z=0} + f_{\text{cylinder}} \rightarrow \min \quad (4.6)$$

4.2. Parameter Settings for Optimization Algorithms

The parameters for LMA are set as follows:

The maximum number of function evaluation: 10^4

The maximum number of iteration: 3×10^3

Step tolerance: 10^{-6}

4.3. Calibration Results and Discussion

We have employed LMA to identify the error parameters of the PCMA. The identified error parameters obtained without data processing (according to [27]) and with data processing are listed in Tables 4 and 5, respectively.

Table 4 Identified error parameters obtained without data processing (according to [27])

No.	$\delta\theta_i$ (deg)	δd_i (mm)	$\delta\alpha_i$ (deg)	δa_i (mm)	$\delta\beta_i$ (deg)
1	×	×	0.1362	− 1.3302	− 0.0206
2	0.0884	−	− 0.2457	− 0.7550	0.7290
3	0.4119	−	− 0.4522	1.3330	− 0.3731
4	0.7237	1.0511	0.8240	×	−
5	0.0142	− 1.5479	×	0.4259	−

Table 5 Identified error parameters obtained with data processing

No.	$\delta\theta_i$ (deg)	δd_i (mm)	$\delta\alpha_i$ (deg)	δa_i (mm)	$\delta\beta_i$ (deg)
1	×	×	0.1382	− 0.9619	− 0.0573
2	0.0707	−	− 0.2059	− 0.4514	0.6679
3	0.6189	−	− 0.3798	0.5518	− 0.1710
4	0.6495	0.3233	0.7696	×	−
5	− 0.1382	0.0264	−	0.4672	−

The maximum of deviation ($\delta_{\max}$) and two times the root-mean-square deviation (2RMS) before and after calibration are listed in Tables 6 and 7, respectively. These tables show the single-point repeatability error of the machine is reduced 1.1 times and the distance accuracy of the machine is enhanced 1.17 times, via the data processing.

Figure 8 shows the curves of twice the standard deviation with and without calibrating of the origin coordinate at

9 default positions. The results of each calibration test are shown in Table 8, 9, and 10. Figure 9 shows volumetric calibration test results.

The results show that all precision indices have increased significantly after calibration. In particular, the calibration results after data processing proposed in this work show that the accuracy of the machine is higher than when the data processing is not performed. Single-point repeatability error is remarkably reduced after calibration for the origin coordinate (Fig. 8). However, the effect of distance accuracy was not obvious in this experiment.

4.4. Evaluation of Measurement Uncertainty

In this subsection, the Type-A measurement uncertainty is evaluated based on data obtained in the calibration test.

The number of locations for each artifact and the number of captured points per each location are listed in Table 8.

The Type-A uncertainty of each item of (4.6) is evaluated by the following equation.

$$u_A = \sqrt{\frac{\sum_{i=1}^n (s_i - \bar{s})^2}{(n-1)}} \quad (4.7)$$

Here, s_i —the individual standard deviations at each location, $\bar{s}$ —the mean standard deviation, n —the number of measurements in a set.

Through the evaluation of Type-A uncertainty of each item, the effect of each item on the accuracy of measuring

Table 6 Maximum of deviation ($\delta_{\max}$) before calibration and after calibration

Artifact class	Before Calib. (mm)		After Calib. (mm)	
	Without data processing	With data processing	1	2
Single-point	11.9970	10.2735	0.0832	0.0729
Sphere	6.2438	6.1258	0.1041	0.0872
Ball bar	7.6732	7.3256	0.1120	0.0979
$z = 0$ plane	3.2830	3.2830	0.1017	0.1182
Cylinder	2.4134	2.4134	0.1131	0.1206

Table 7 Two times the root-mean-square deviation (2RMS) of before and after calibration

Artifact class	Before Calib. (mm)		After Calib. (mm)	
	Without data processing	With data processing	1	2
Single-point	10.3574	10.6030	0.0847	0.0776
Sphere	6.2388	6.5146	0.1063	0.0885
Ball bar	7.5635	7.6134	0.1145	0.0981
$z = 0$ plane	3.3451	3.3451	0.1021	0.1037
cylinder	2.4058	2.4058	0.1138	0.1213

Fig. 8 The twice the standard deviation $2S^{\text{point}}$ at 9 default locations

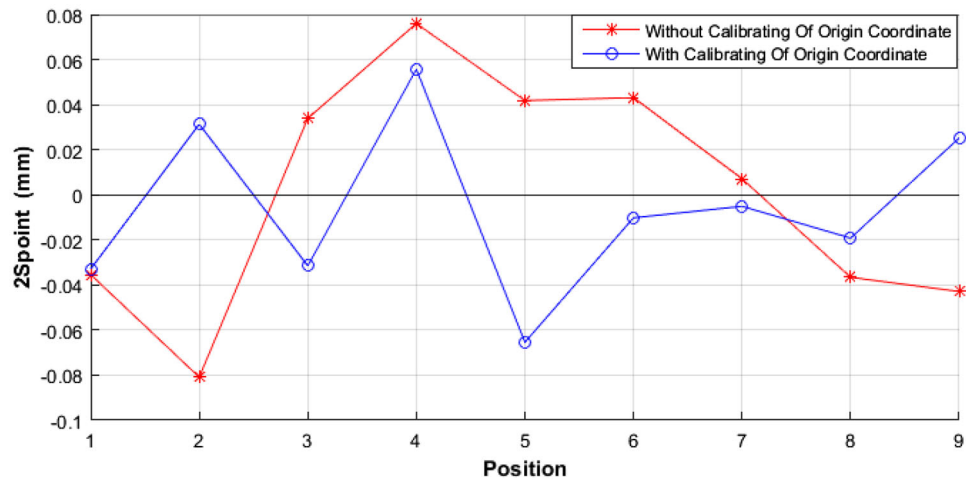


Table 8 The number of positions for each artifact, and the number of captured points per each position

Artifact	Calibration data	
	Locations	Points per location
Chamfered hole	10	10
Sphere	10	10
Ball bar	10	10
$z = 0$ plane	1	10
Cylinder	1	10

machine could be evaluated. As a result, the error of volumetric calibration test is of the greatest influence having 43% of total error. So, the total error could be reduced by taking enough tests on the volumetric calibration (Tables 11, 12, and 13).

5. Conclusion

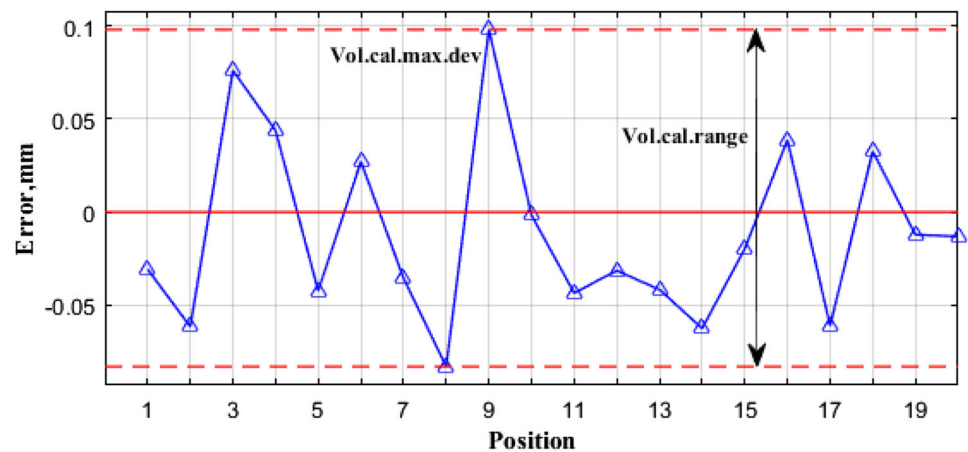
In this paper, a kinematic calibration approach has been proposed for 5DOF PCMA. In the calibration, ASME B89.4.22-2004 has been referenced, and for volumetric calibration test, a self-made ball bar has been employed. Data processing for the exclusion of rough error which reduces the accuracy of measuring machine has been presented, and the machine coordinate origin determination has been carried out for the improvement in the accuracy. An objective function for the identification of kinematic error parameters has been derived, and its optimized solution has been obtained by using Levenberg–Marquardt Algorithm (LMA). Through the evaluation of Type-A uncertainty, the effect of each calibration item on the accuracy of measuring machine has been evaluated. A calibration experiment has been conducted, and the result

Table 9 Type-A uncertainty of single-point performance test (f_{point})

Position no.	1	2	3	4	5	6	7	8	9	10
s_i (mm)	0.0150	0.0253	0.0121	0.0329	0.0213	0.0137	0.0246	0.0218	0.0151	0.0232
Mean (mm)	0.0205									
Max (mm)	0.0329									
Min (mm)	0.0121									
uA	0.0065									

Table 10 Type-A uncertainty of effective diameter calibration test (f_{sphere})

Position no.	1	2	3	4	5	6	7	8	9	10
$D_i - D_{\text{cal}}$ (mm)	0.0271	-0.0136	0.0513	0.0318	0.0426	0.0426	0.0326	0.0274	-0.0214	0.0326
D_{max} (mm)	0.0513									
Mean (mm)	0.0253									
uA	0.0238									

Fig. 9 Volumetric calibration test results**Table 11** Type-A uncertainty of volumetric calibration test (f_{length}), (unit:mm)

Position no.	1	2	3	4	5	6	7	8	9	10
$L_i - L_{\text{cal}}$ (mm)	- 0.0217	- 0.0523	0.0505	0.0425	- 0.0516	0.0235	- 0.0363	- 0.0459	0.0536	- 0.0358
Mean (mm)	- 0.0074									
Max (mm)	0.0536									
range (mm)	0.1059									
uA	0.0445									

Table 12 Type-A uncertainty of base plane $z = 0$ ($f_{z=0}$)

Position no.	1	2	3	4	5	6	7	8	9	10
z_i (mm)	0.0127	0.0135	0.0023	- 0.0035	0.0154	0.0137	- 0.0162	0.0185	0.0162	- 0.0147
Mean (mm)	0.0058									
uA	0.0131									

Table 13 Type-A uncertainty of base cylinder $x^2 + y^2 = R^2$ (f_{cylinder})

Position no.	1	2	3	4	5	6	7	8	9	10
${}^{\text{cyl}}\delta_i$ (mm)	- 0.0132	0.0152	- 0.0103	0.0125	0.0201	0.0185	0.0153	- 0.0159	- 0.0129	0.0193
Mean (mm)	0.0049									
uA	0.0156									

shows the repeatability has been enhanced 1.1 times and the distance accuracy is enhanced 1.17 times, which validates the accuracy of the proposed calibration method.

The methods proposed in this work will contribute to the improvement in calibration method of PCMA.

Authors' Contributions CHR and CMR conceived the mathematical models, interpreted the computational results, and wrote the paper. JGK has carried out the calibration experiment and conducted the evaluation test of measurement uncertainty. GC implemented and performed most of the simulations and optimization calculations in consultation with JSP. All authors gave final approval for publication.

Funding This work was supported by National Natural Science Foundation of China (Grant No. 51075095).

Compliance with Ethical Standards

Conflict of interest We have no competing interests.

Ethical Standards This work did not involve any active collection of human data, but only computer simulations of human behavior.

Data Accessibility Statement This work does not have any experimental data.

References

- [1] T. Messay, R. Ordóñez and E. Marcil, Computationally efficient and robust kinematic calibration methodologies and their application to industrial robots. *Robotics and Computer-Integrated Manufacturing*, **37** (2016) 33–48.
- [2] J. Santolaria, A.C. Majarena and D. Samper, et al., Articulated arm coordinate measuring machine calibration by laser tracker multilateration. *The Scientific World Journal*, (2014) 681853.
- [3] R. He, Y. Zhao and S. Yang, et al. Kinematic-parameter identification for serial-robot calibration based on POE formula. *Robotics IEEE Transactions on*, **26**(3) (2010) 411–423.
- [4] I.M. Chen and G. Yang, Kinematic calibration of modular reconfigurable robots using product-of-exponentials formula. *Journal of Robotic Systems*, **14**(11) (1997) 807.
- [5] S. K. Mustafa, B.L. Wen and G. Yang, et al., Cable-driven robots. Springer London (2015).
- [6] I.M. Chen, G. Yang and C.T. Tan, et al., Local POE model for robot kinematic calibration. *Mechanism & Machine Theory*, **36**(11–12) (2001) 1215–1239.
- [7] W. Zhang, X. Ma and L. Cui, et al., 3 points calibration method of part coordinates for arc welding robot. *Intelligent robotics and applications*. Springer Berlin Heidelberg, (2008) 216–224.
- [8] S. Hayati and M. Mirmirani, Improving the absolute positioning accuracy of robot manipulators. *Robotic Syst.*, **2** (1985) 397–413.
- [9] I.M. Chen, G. Yang and C.T. Tan, et al., Local POE model for robot kinematic calibration. *Mechanism & Machine Theory*, **36**(11–12) (2001) 1215–1239.
- [10] G. Calafiore, M. Indri and B. Bona, Robot dynamic calibration: Optimal excitation trajectories and experimental parameter estimation. *Journal of Robotic Systems*, **18**(2) (2001) 55–68.
- [11] A. Joubair and I.A. Bonev, Kinematic calibration of a six-axis serial robot using distance and sphere constraints. *The International Journal of Advanced Manufacturing Technology*, **77**(1) (2015) 515–523.
- [12] A. Li, Z. Ma and D. Wu, Hand-eye calibration for robot measuring system based on adaptive differential evolution. *Intelligent control and automation*. IEEE, (2010) 3241–3245.
- [13] H. Zhuang, L.K. Wang and Z.S. Roth, Error-model-based robot calibration using a modified CPC model. *Robotics and Computer-Integrated Manufacturing*, **10**(4) (1993) 287–299.
- [14] S. Marie, E. Courteille and P. Maurine, Elasto-geometrical modeling and calibration of robot manipulators: Application to machining and forming applications. *Mechanism & Machine Theory*, **69**(6) (2013) 13–43.
- [15] G. Schreiber, A. Albu-Schaefer and N. Sporer, Method for calibrating a robot arm:, EP1302285. (2003).
- [16] M. Crenganis, R. Breaz and G. Racu, et al., Kinematic solutions of a 7 DOF robotic arm using redundancy circle and fuzzy models. *Applied Mechanics & Materials*, **555** (2014) 320–326.
- [17] Z.S. Roth, B.W. Mooring and B. Ravani, An overview of robot calibration. *Information Technology Journal*, **3**(5) (1987) 377–385.
- [18] Y. Bai, On the comparison of model-based and modelless robotic calibration based on a fuzzy interpolation method. *The International Journal of Advanced Manufacturing Technology*, **31**(11) (2007) 1243–1250.
- [19] PN EN ISO 10360, Specyfikacje Geometrii Wyrobów (GPS) Badania odbiorcze i okresowe współrzędnościowych maszyn pomiarowych (CMM) (Geometrical Product Specifications (GPS)—Acceptance and reverification tests for coordinate measuring machines (CMM)).
- [20] VDI/VDE 2617 Genauigkeit von Koordinatenmessgeräten Kenngrößen und deren Prüfung.
- [21] V.C. Nardelli and G.D. Donatelli, A simple solution to interim check of coordinate measuring machines, in: XVIII IMEKO world congress, Rio de Janeiro, Brazil, (2006).
- [22] S.A. Farooqui, E.P. Morse, Alternative artifacts for evaluating scanning CMM performance, in: ASPE annual meeting, Orlando, USA, (2004).
- [23] L. De Chiffre, H.E. Hansen and R.E. Morace, Comparison of coordinate measuring machines using an optomechanical hole plate, *CIRP Annals—Manufacturing Technology* **54** (2005) 479–482.
- [24] B. Bringmann and A. Küng, A measuring artifact for true 3D machine testing and calibration, *CIRP Annals—Manufacturing Technology*, **54** (2005) 471–474.
- [25] I. Kovac and A. Frank, Testing and calibration of coordinate measuring arms, *Precision Engineering*, **25** (2001) 90–99.
- [26] K. Shimojima, R. Furutani, K. Takamasu and K. Araki, The estimation method of uncertainty of articulated coordinate measuring machine, in: XVII IMEKO world congress, Dubrovnik, Croatia, (2003).
- [27] ASME B89.4.22-2004, Methods for performance evaluation of articulated arm coordinate measuring machines (CMM), New York (2004).
- [28] J. Li, L.D. Yu and J.Q. Sun, et al. A kinematic model for parallel-joint coordinate measuring machine. *Journal of Mechanisms and Robotics*, **5**(4) (2013) 044501–044501-4.
- [29] Y. Guo, S.B. Yin and Y.J. Ren, et al. A multilevel calibration technique for an industrial robot with parallelogram mechanism. *Precision Engineering*, **40** (2015) 261–272.
- [30] J. Ziegert and P. Datseris, Basic considerations for robot calibration. *Proceedings of the IEEE conference on robotics and automation*, New York, USA, March, (1988) 932–938.
- [31] H.N. Nguyen, J. Zhou and H.J. Kang, A calibration method for enhancing robot accuracy through integration of an extended Kalman filter algorithm and an artificial neural network. *Neurocomputing*, **151** (2015) 996–1005.
- [32] D. Marquardt, An algorithm for least-squares estimation of nonlinear parameters. *SIAM Journal on Applied Mathematics*, **11**(2) (1963).